

Question Number	Answer	Additional Guidance	Mark
2 ai	• calculation of mean distance (1) • correct use of r^2 (1) • correct answer (1)	Allow ECF for MP2 and MP3 $L = 4.4 \text{ (mm)}$ $r^2 = 0.01 \text{ (mm}^2\text{)}$ $0.14 \text{ (mm}^3 \text{ min}^{-1}\text{)}$ 0.138 gets 2 marks correct answer with no working gains (3) marks	3

Question Number	Answer	Additional Guidance	Mark
2aii	<u>volume of carbon dioxide produced</u> (1) volume of oxygen used - RQ value calculated (1)	$0.11 \div 0.14$ $0.786/0.79 / 0.8/0.80$ Allow ecf of wrong rate of oxygen from 2ai $0.11 \div 0.138 = 0.797 / 0.80$ $0.11 \div 0.55 = 0.20$	2

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2aiii	soda lime absorbs carbon dioxide so must be removed so carbon dioxide produced can be measured.	Accept glass beads do not absorb carbon dioxide so carbon dioxide produced can be measured Ignore ref to measuring RQ	1

Question Number	Answer	Additional Guidance	Mark
2bi	An answer that includes two of the following: - mass of maggot (1) - species (1) - age (1)	One abiotic and two biotic = 1mk two abiotic = 0 Accept source Ignore size, sex, type, genetic variation, microbes in tube, competition, disease	2

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2bii	A description that includes the following: • variable with suitable control method described (1) • results are not valid / description of expected effect on the dependent variable (1)	Must give detail not just eg use the same mass Must be directional if giving expected effect	